

Mental Health Survey: Exploring the Attitudes and Perceptions of Persons who Underwent Mental Health Treatment

Shardha Ramlal

2026-06-14

Contents

Introduction	2
Data Description	2
Exploratory Data Analysis	3
Data Quality - Completeness	4
How does missing data look according to treatment?	5
Data Cleaning	5
Summary Statistics of dataset: Demographics by Treatment	5
Summary Statistics of dataset: Workplace Attitudes by Treatment	6
Geographic Distribution of Responses	10
Logistic Regression Analysis of Factors affecting of Mental Health Treatment Behaviour	10
Forest Plot	13
Appendix	14
Data Cleaning	14
p values double check gtsummary table	15
Logistic Regression Raw Output	18
References	19

Introduction

Data link & Research Inspiration: <https://www.kaggle.com/datasets/osmi/mental-health-in-tech-survey1>

The data was from a 2014 survey that measured attitudes towards mental health as well as frequency of mental health disorders in the workforce, specifically tech workplaces.

Research Question: What are the strongest predictors of mental health illness or certain attitudes towards mental health in the workplace?

Data Description

Variables include:

- Timestamp
- Age
- Gender
- Country
- state: If you live in the United States, which state or territory do you live in?
- self_employed: Are you self-employed?
- family_history: Do you have a family history of mental illness?
- treatment: Have you sought treatment for a mental health condition?
- work_interfere: If you have a mental health condition, do you feel that it interferes with your work?
- no_employees: How many employees does your company or organization have?
- remote_work: Do you work remotely (outside of an office) at least 50% of the time?
- tech_company: Is your employer primarily a tech company/organization?
- benefits: Does your employer provide mental health benefits?
- care_options: Do you know the options for mental health care your employer provides?
- wellness_program: Has your employer ever discussed mental health as part of an employee wellness program?
- seek_help: Does your employer provide resources to learn more about mental health issues and how to seek help?
- anonymity: Is your anonymity protected if you choose to take advantage of mental health or substance abuse treatment resources?
- leave: How easy is it for you to take medical leave for a mental health condition?
- mental_health_consequence: Do you think that discussing a mental health issue with your employer would have negative consequences?
- phys_health_consequence: Do you think that discussing a physical health issue with your employer would have negative consequences?
- coworkers: Would you be willing to discuss a mental health issue with your coworkers?

- supervisor: Would you be willing to discuss a mental health issue with your direct supervisor(s)?
- mental_health_interview: Would you bring up a mental health issue with a potential employer in an interview?
- phys_health_interview: Would you bring up a physical health issue with a potential employer in an interview?
- mental_vs_physical: Do you feel that your employer takes mental health as seriously as physical health?
- obs_consequence: Have you heard of or observed negative consequences for coworkers with mental health conditions in your workplace?
- comments: Any additional notes or comments

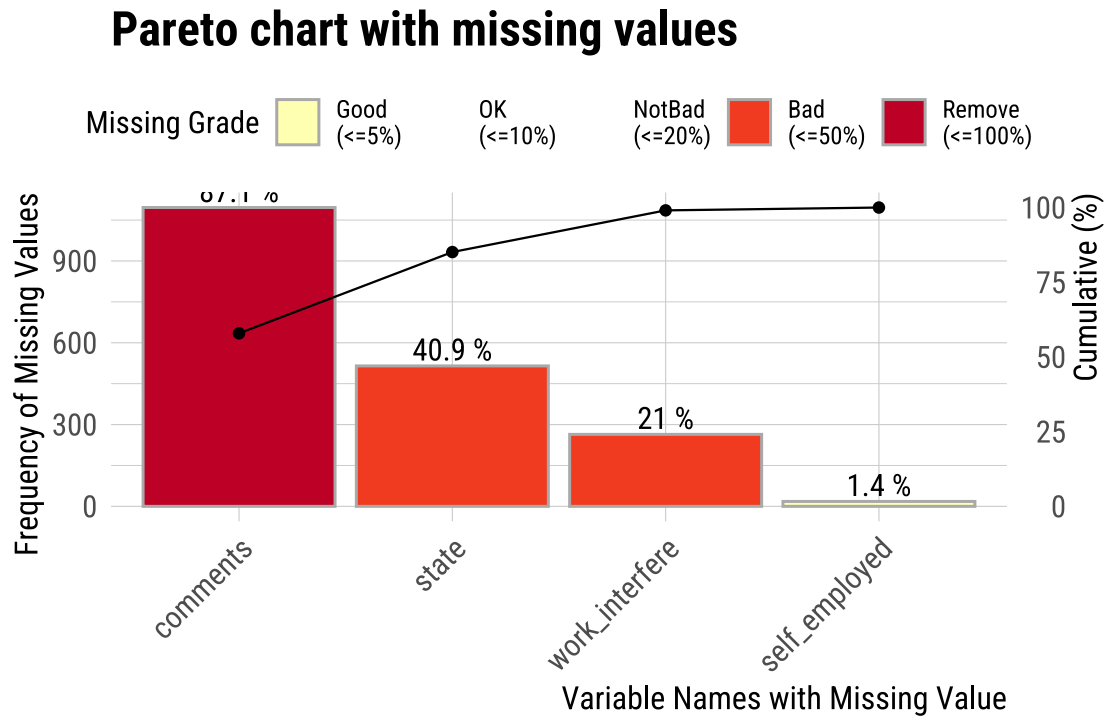
Exploratory Data Analysis

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Age	0	1	79428148	2818299443	-1726	27	31	36	9999999999	

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
Gender	0	1.0000000	FALSE	47	Mal: 618, mal: 206, Fem: 123, M: 116
Country	0	1.0000000	FALSE	48	Uni: 751, Uni: 185, Can: 72, Ger: 45
state	515	0.5909452	FALSE	45	CA: 138, WA: 70, NY: 57, TN: 45
self_employed	18	0.9857029	FALSE	2	No: 1095, Yes: 146
family_history	0	1.0000000	FALSE	2	No: 767, Yes: 492
treatment	0	1.0000000	FALSE	2	Yes: 637, No: 622
work_interfere	264	0.7903098	FALSE	4	Som: 465, Nev: 213, Rar: 173, Oft: 144
no_employees	0	1.0000000	FALSE	6	6-2: 290, 26-: 289, Mor: 282, 100: 176
remote_work	0	1.0000000	FALSE	2	No: 883, Yes: 376
tech_company	0	1.0000000	FALSE	2	Yes: 1031, No: 228
benefits	0	1.0000000	FALSE	3	Yes: 477, Don: 408, No: 374
care_options	0	1.0000000	FALSE	3	No: 501, Yes: 444, Not: 314
wellness_program	0	1.0000000	FALSE	3	No: 842, Yes: 229, Don: 188
seek_help	0	1.0000000	FALSE	3	No: 646, Don: 363, Yes: 250
anonymity	0	1.0000000	FALSE	3	Don: 819, Yes: 375, No: 65
leave	0	1.0000000	FALSE	5	Don: 563, Som: 266, Ver: 206, Som: 126
mental_health_consequence	0	1.0000000	FALSE	3	No: 490, May: 477, Yes: 292
phys_health_consequence	0	1.0000000	FALSE	3	No: 925, May: 273, Yes: 61
coworkers	0	1.0000000	FALSE	3	Som: 774, No: 260, Yes: 225
supervisor	0	1.0000000	FALSE	3	Yes: 516, No: 393, Som: 350
mental_health_interview	0	1.0000000	FALSE	3	No: 1008, May: 207, Yes: 44
phys_health_interview	0	1.0000000	FALSE	3	May: 557, No: 500, Yes: 202
mental_vs_physical	0	1.0000000	FALSE	3	Don: 576, Yes: 343, No: 340

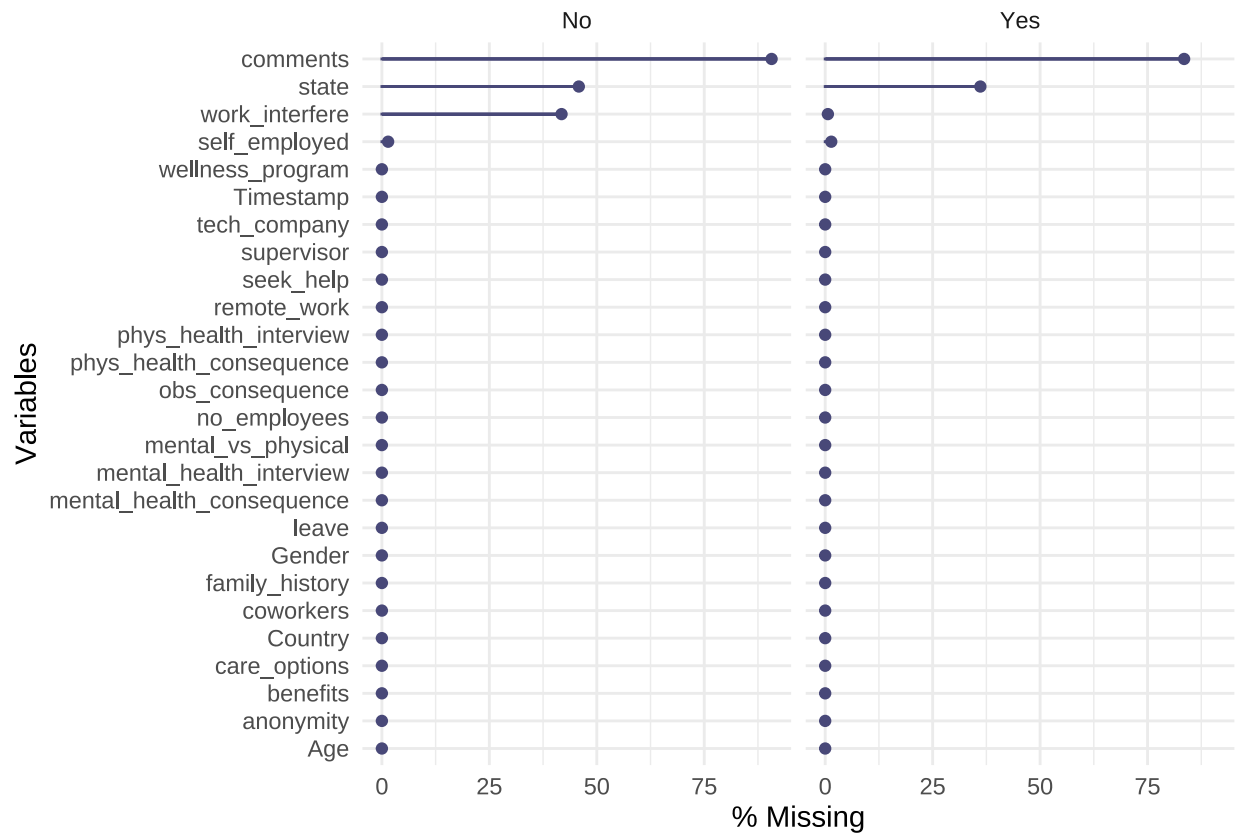
skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
obs_consequence	0	1.0000000	FALSE	2	No: 1075, Yes: 184
comments	1096	0.1294678	FALSE	159	* S: 5, -: 1, (ye: 1, A c: 1

Data Quality - Completeness



Missing data seemed to be concentrated on the work_interfere and self_employed variable. The state variable was only applicable for the USA and the comments variable was optional.

How does missing data look according to treatment?



Persons who did not seek mental health treatment before mostly did not appear to have a mental health condition.

Data Cleaning

We observed there was further cleaning to be done with the dataset:

- There were outliers in Age (value of 329), negative values (-1726, -29, -1), and persons 5, 8 and 11 years old which were assumed these are errors and the legal working age in the dataset was 18 and over
- discrepancies in the coding of Gender - f, female, m, male, femake etc.

Summary Statistics of dataset: Demographics by Treatment

Characteristic	Treatment for Mental Health			p-value
	Overall N = 1,259 ¹	No N = 622 ¹	Yes N = 637 ¹	
Age	32.1 (7.3)	31.5 (7.0)	32.6 (7.6)	0.014²

Characteristic	Treatment for Mental Health			p-value
	Overall N = 1,259 ¹	No N = 622 ¹	Yes N = 637 ¹	
Gender				<0.001³
Female	247 (19.9%)	77 (12.5%)	170 (27.2%)	
Male	989 (79.6%)	540 (87.4%)	449 (71.8%)	
Other	7 (0.6%)	1 (0.2%)	6 (1.0%)	
Self Employed				0.468⁴
No	1,095 (88.2%)	545 (88.9%)	550 (87.6%)	
Yes	146 (11.8%)	68 (11.1%)	78 (12.4%)	
Family History of Mental Illness				<0.001⁴
No	767 (60.9%)	495 (79.6%)	272 (42.7%)	
Yes	492 (39.1%)	127 (20.4%)	365 (57.3%)	
Interfere with Ability to Work				<0.001⁴
Never	213 (16.9%)	183 (29.4%)	30 (4.7%)	
Not applicable	264 (21.0%)	260 (41.8%)	4 (0.6%)	
Often	144 (11.4%)	21 (3.4%)	123 (19.3%)	
Rarely	173 (13.7%)	51 (8.2%)	122 (19.2%)	
Sometimes	465 (36.9%)	107 (17.2%)	358 (56.2%)	

¹Mean (SD); n (%)

²Wilcoxon rank sum test

³Fisher's exact test

⁴Pearson's Chi-squared test

There appear to be significant associations between treatment and Age, Gender, Family History of Mental Illness and whether having a mental health condition affected work.

Summary Statistics of dataset: Workplace Attitudes by Treatment

Characteristic	Treatment for Mental Health			p-value
	Overall N = 1,259 ¹	No N = 622 ¹	Yes N = 637 ¹	
Number of Employees				0.119²
1-5	162 (12.9%)	71 (11.4%)	91 (14.3%)	
6-25	290 (23.0%)	162 (26.0%)	128 (20.1%)	
26-100	289 (23.0%)	139 (22.3%)	150 (23.5%)	
100-500	176 (14.0%)	81 (13.0%)	95 (14.9%)	

Treatment for Mental Health				
Characteristic	Overall N = 1,259 ¹	No N = 622 ¹	Yes N = 637 ¹	p-value
500-1000	60 (4.8%)	33 (5.3%)	27 (4.2%)	0.339 ²
More than 1000	282 (22.4%)	136 (21.9%)	146 (22.9%)	
Remote Work				
No	883 (70.1%)	444 (71.4%)	439 (68.9%)	0.263 ²
Yes	376 (29.9%)	178 (28.6%)	198 (31.1%)	
Tech Company				
No	228 (18.1%)	105 (16.9%)	123 (19.3%)	<0.001 ²
Yes	1,031 (81.9%)	517 (83.1%)	514 (80.7%)	
Benefits				
Don't know	408 (32.4%)	257 (41.3%)	151 (23.7%)	<0.001 ²
No	374 (29.7%)	193 (31.0%)	181 (28.4%)	
Yes	477 (37.9%)	172 (27.7%)	305 (47.9%)	
Care Options				<0.001 ²
No	501 (39.8%)	294 (47.3%)	207 (32.5%)	
Not sure	314 (24.9%)	191 (30.7%)	123 (19.3%)	
Yes	444 (35.3%)	137 (22.0%)	307 (48.2%)	0.003 ²
Wellness Program				
Don't know	188 (14.9%)	107 (17.2%)	81 (12.7%)	
No	842 (66.9%)	422 (67.8%)	420 (65.9%)	0.004 ²
Yes	229 (18.2%)	93 (15.0%)	136 (21.4%)	
Seek Help				
Don't know	363 (28.8%)	197 (31.7%)	166 (26.1%)	<0.001 ²
No	646 (51.3%)	323 (51.9%)	323 (50.7%)	
Yes	250 (19.9%)	102 (16.4%)	148 (23.2%)	
Anonymity				<0.001 ²
Don't know	819 (65.1%)	448 (72.0%)	371 (58.2%)	
No	65 (5.2%)	27 (4.3%)	38 (6.0%)	
Yes	375 (29.8%)	147 (23.6%)	228 (35.8%)	<0.001 ²
Leave				
Don't know	563 (44.7%)	309 (49.7%)	254 (39.9%)	
Somewhat difficult	126 (10.0%)	44 (7.1%)	82 (12.9%)	
Somewhat easy	266 (21.1%)	135 (21.7%)	131 (20.6%)	
Very difficult	98 (7.8%)	31 (5.0%)	67 (10.5%)	

Characteristic	Treatment for Mental Health			p-value
	Overall N = 1,259 ¹	No N = 622 ¹	Yes N = 637 ¹	
Very easy	206 (16.4%)	103 (16.6%)	103 (16.2%)	<0.001 ²
Mental Health Consequence				
Maybe	477 (37.9%)	224 (36.0%)	253 (39.7%)	
No	490 (38.9%)	280 (45.0%)	210 (33.0%)	
Yes	292 (23.2%)	118 (19.0%)	174 (27.3%)	0.185 ²
Physical Health Consequence				
Maybe	273 (21.7%)	127 (20.4%)	146 (22.9%)	
No	925 (73.5%)	470 (75.6%)	455 (71.4%)	
Yes	61 (4.8%)	25 (4.0%)	36 (5.7%)	0.050 ²
Coworkers				
No	260 (20.7%)	141 (22.7%)	119 (18.7%)	
Some of them	774 (61.5%)	384 (61.7%)	390 (61.2%)	
Yes	225 (17.9%)	97 (15.6%)	128 (20.1%)	0.422 ²
Supervisor				
No	393 (31.2%)	186 (29.9%)	207 (32.5%)	
Some of them	350 (27.8%)	170 (27.3%)	180 (28.3%)	
Yes	516 (41.0%)	266 (42.8%)	250 (39.2%)	0.002 ²
Mental Health Interview				
Maybe	207 (16.4%)	125 (20.1%)	82 (12.9%)	
No	1,008 (80.1%)	479 (77.0%)	529 (83.0%)	
Yes	44 (3.5%)	18 (2.9%)	26 (4.1%)	0.183 ²
Physical Health Interview				
Maybe	557 (44.2%)	290 (46.6%)	267 (41.9%)	
No	500 (39.7%)	241 (38.7%)	259 (40.7%)	
Yes	202 (16.0%)	91 (14.6%)	111 (17.4%)	<0.001 ²
Mental vs Physical				
Don't know	576 (45.8%)	316 (50.8%)	260 (40.8%)	
No	340 (27.0%)	138 (22.2%)	202 (31.7%)	
Yes	343 (27.2%)	168 (27.0%)	175 (27.5%)	<0.001 ²
Observed Negative Consequence				
No	1,075 (85.4%)	566 (91.0%)	509 (79.9%)	
Yes	184 (14.6%)	56 (9.0%)	128 (20.1%)	

Characteristic	Treatment for Mental Health			p-value
	Overall N = 1,259 ¹	No N = 622 ¹	Yes N = 637 ¹	

¹n (%)

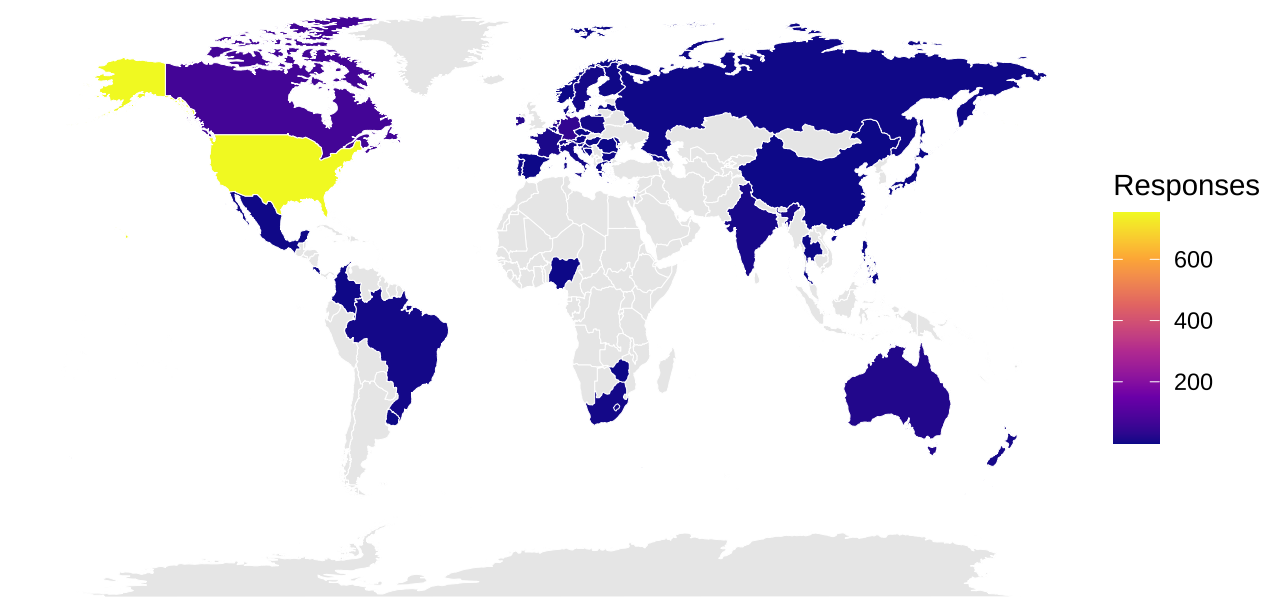
²Pearson's Chi-squared test

There also seemed to be significant associations between treatment and:

- workplace provided mental health benefits,
- knew mental health care options in the workplace
- employer discussed mental health as part of a wellness program
- employer provided resources to learn more about mental health
- anonymity protected if using mental health services
- how easy it was to take mental health leave
- discussed mental health issues with employer had negative consequences
- willing to discuss mental health issues with coworkers
- willing to bring up mental health issues with a potential employer
- see employer taking mental health as seriously as physical health
- heard / observed negative consequences for coworkers with mental health conditions

Geographic Distribution of Responses

Global Distribution of Survey Responses



Most respondents geographical disposition was the USA.

Logistic Regression Analysis of Factors affecting of Mental Health Treatment Behaviour

Demographic, Perception and Attitude factors were modeled against treatment using a logistic regression model.

Characteristic	OR	95% CI	p-value
(Intercept)	0.09	0.02, 0.33	<0.001
Age (centred)	1.03	1.00, 1.06	0.019
Gender			
Female	—	—	
Male	0.47	0.29, 0.74	0.002
Other	1.14	0.14, 25.0	0.914
Self Employed			
No	—	—	

Characteristic	OR	95% CI	p-value
Yes	0.83	0.43, 1.58	0.564
Family History of Mental Illness			
No	—	—	
Yes	2.58	1.80, 3.71	<0.001
Interfere with Ability to Work			
Never	—	—	
Not applicable	0.09	0.03, 0.24	<0.001
Often	33.6	17.0, 69.2	<0.001
Rarely	12.6	7.10, 22.8	<0.001
Sometimes	18.2	11.0, 31.1	<0.001
Number of Employees			
1-5	—	—	
6-25	1.18	0.61, 2.26	0.618
26-100	1.42	0.71, 2.84	0.326
100-500	1.34	0.62, 2.89	0.461
500-1000	1.34	0.46, 3.94	0.594
More than 1000	1.05	0.48, 2.25	0.909
Remote Work			
No	—	—	
Yes	0.87	0.58, 1.31	0.505
Tech Company			
No	—	—	
Yes	0.94	0.58, 1.51	0.807
Benefits			
Don't know	—	—	
No	1.20	0.70, 2.03	0.509
Yes	1.99	1.18, 3.40	0.011
Care Options			
No	—	—	
Not sure	0.80	0.50, 1.29	0.359
Yes	2.03	1.27, 3.27	0.003
Wellness Program			
Don't know	—	—	
No	1.01	0.55, 1.86	0.970
Yes	0.72	0.35, 1.46	0.364

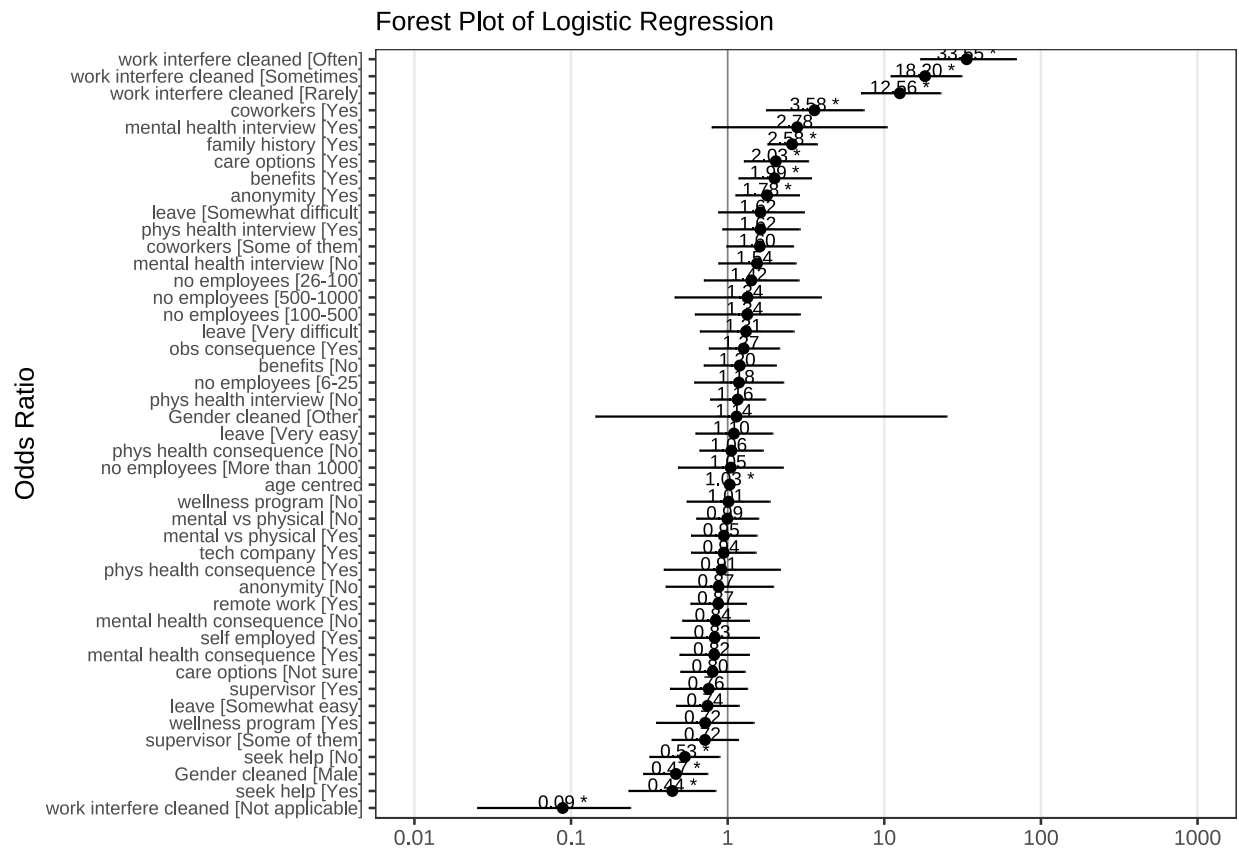
Characteristic	OR	95% CI	p-value
Seek Help			
Don't know	—	—	
No	0.53	0.32, 0.89	0.016
Yes	0.44	0.23, 0.84	0.012
Anonymity			
Don't know	—	—	
No	0.87	0.40, 1.95	0.739
Yes	1.78	1.12, 2.85	0.015
Leave			
Don't know	—	—	
Somewhat difficult	1.62	0.87, 3.07	0.132
Somewhat easy	0.74	0.47, 1.18	0.206
Very difficult	1.31	0.67, 2.64	0.442
Very easy	1.10	0.62, 1.93	0.749
Mental Health Consequence			
Maybe	—	—	
No	0.84	0.51, 1.37	0.480
Yes	0.82	0.49, 1.37	0.445
Physical Health Consequence			
Maybe	—	—	
No	1.06	0.66, 1.68	0.820
Yes	0.91	0.39, 2.16	0.832
Coworkers			
No	—	—	
Some of them	1.60	0.98, 2.62	0.058
Yes	3.58	1.76, 7.38	<0.001
Supervisor			
No	—	—	
Some of them	0.72	0.44, 1.17	0.180
Yes	0.76	0.43, 1.33	0.334
Mental Health Interview			
Maybe	—	—	
No	1.54	0.87, 2.72	0.135
Yes	2.78	0.79, 10.4	0.120
Physical Health Interview			

Characteristic	OR	95% CI	p-value
Maybe	—	—	
No	1.16	0.77, 1.74	0.476
Yes	1.62	0.93, 2.88	0.095
Mental vs Physical			
Don't know	—	—	
No	1.0	0.63, 1.57	0.983
Yes	0.95	0.58, 1.53	0.819
Observed Negative Consequence			
No	—	—	
Yes	1.27	0.76, 2.14	0.372

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Null deviance = 1,691; Null df = 1,219; Log-likelihood = -432; AIC = 957;
BIC = 1,192; Deviance = 865; Residual df = 1,174; No. Obs. = 1,220

Forest Plot



A binary, logistic regression model was used to examine factors affecting mental health treatment seeking

behaviour among 1220 respondents. This model used the logit link function and used the dependent variable of treatment, while using all demographic, attitudes and perception variables as independent variables.

The residual deviance was 864.61 on 1174 degrees of freedom yielded $p=1.000$, which implied we do not reject the null hypothesis and the model was satisfactory. The model AIC was 957. Significant predictors of seeking mental health treatment in the workplace included demographic, workplace attitudes and perceptions.

Demographic factors such as age, gender, family history of mental illness and interfered with ability to work were significant predictors in the model. Every one year increase from the mean age was associated with a 3% increase in odds of seeking treatment (OR=1.03, CI: 1.00, 1.06, $p=0.019$). Male respondents had a 53% odds lower treatment than the reference level of females, while holding the rest of predictors constant (OR=0.47, CI: 0.29, 0.74, $p=0.002$). Other respondents had 1.14 higher odds of treatment than females (OR=1.14, CI: 0.14, 25.0, $p=0.914$), however the confidence interval is large, implying possible few observations, which was a limitation.

Having a family history of mental health illness, yielded 2.58 times higher odds of seeking treatment (OR=2.58, CI: 1.80, 3.71, $p<0.001$), while respondents reporting their condition relative to never interfering with their work showed: those whose condition often interfered with their work reported 33.6 times higher odds (OR=33.6, CI: 17.0, 69.2, $p<0.001$), those who reported sometimes interfered with their work showed 18.2 times higher odds (OR=18.2, CI: 11.0, 31.1, $p<0.001$), those who reported rarely showed 12.6 higher odds (OR=12.6, CI: 7.10, 22.8, $p<0.001$), while those who reported not applicable showed 91% lower odds than those who never reporting experiencing affecting their work (OR=0.09, CI: 11.0, 31.1, $p<0.001$).

The workplace and attitude factors that were significantly significant included employer provided mental health benefits, if employees knew their care options for mental health in the workplace, employer provided resources to learn more about mental health, anonymity protected if using mental health services, discussed mental health issues with employer had negative consequences and willing to discuss mental health issues with coworkers.

Having mental health benefits (OR=1.99, CI: 1.18, 3.40, $p=0.011$) and caving care options (OR=2.03, CI: 1.27, 3.27, $p=0.003$) were associated with double the odds of treatment seeking behaviour. Those whose employer provided resources to learn more about mental health reported 56% lower odds in companion to those who did not know (OR=0.44, CI: 0.23, 0.84, $p=0.012$), while those whose employees did not provide resources however a 47% lower odds of seeking mental health treatment compared to those who did not know (OR=0.53, CI: 0.32, 0.89, $p=0.016$).

Having anonymity protected was associated 78% higher odds of seeking mental health treatment (OR=1.78, CI: 1.12, 2.85, $p=0.015$). Those willing to discuss mental health issues with coworkers showed a 3.48 times the odds of seeking treatment compared to those who were not willing (OR=3.58, CI: 1.76, 7.38, $p<0.001$).

How easy it was to take mental health leave did not have an association with mental health treatment after controlling with other factors, conversely was seen when univariate testing. Company size, remote work status and tech company designation were not significant predictors in the model.

Appendix

Data Cleaning

Age	n
329	1
72	1
65	1
62	1
61	1

Age	n
60	2

Age	n
-1726	1
-29	1
-1	1
5	1
8	1
11	1
18	7
19	9
20	6
21	16

Gender	n
A little about you	1
Agender	1
All	1
Androgyne	1
cis-female/femme	1
Cis Female	1
cis male	1
Cis Male	2
Cis Man	1
Enby	1

Gender_cleaned	n
Female	247
Male	989
Other	7
NA	16

p values double check gtsummary table

Table 4: Chi-Square Test Results

variable	statistic	p.value	parameter	method	pct_under_5
Gender_cleaned	46.9227944	0.0000000	2	Pearson's Chi-squared test	33.33333
self_employed	0.5265341	0.4680677	1	Pearson's Chi-squared test	0.00000
family_history	179.8126158	0.0000000	1	Pearson's Chi-squared test	0.00000

variable	statistic	p.value	parameter	method	pct_under_5
work_interfere_cleaned	594.9243179	0.0000000	4	Pearson's Chi-squared test	0.00000
no_employees	8.7648050	0.1188211	5	Pearson's Chi-squared test	0.00000
remote_work	0.9135588	0.3391719	1	Pearson's Chi-squared test	0.00000
tech_company	1.2512464	0.2633146	1	Pearson's Chi-squared test	0.00000
benefits	64.8385903	0.0000000	2	Pearson's Chi-squared test	0.00000
care_options	94.7587268	0.0000000	2	Pearson's Chi-squared test	0.00000
wellness_program	11.4976499	0.0031865	2	Pearson's Chi-squared test	0.00000
seek_help	10.9342218	0.0042234	2	Pearson's Chi-squared test	0.00000
anonymity	26.4218920	0.0000018	2	Pearson's Chi-squared test	0.00000
leave	29.9434966	0.0000050	4	Pearson's Chi-squared test	0.00000
mental_health_consequence	22.3272848	0.0000142	2	Pearson's Chi-squared test	0.00000
phys_health_consequence	3.3709594	0.1853555	2	Pearson's Chi-squared test	0.00000
coworkers	6.0012998	0.0497547	2	Pearson's Chi-squared test	0.00000
supervisor	1.7255074	0.4219984	2	Pearson's Chi-squared test	0.00000
mental_health_interview	12.6901594	0.0017554	2	Pearson's Chi-squared test	0.00000
phys_health_interview	3.3996980	0.1827111	2	Pearson's Chi-squared test	0.00000
mental_vs_physical	17.4581253	0.0001618	2	Pearson's Chi-squared test	0.00000
obs_consequence	31.0219289	0.0000000	1	Pearson's Chi-squared test	0.00000

```
##
## Fisher's Exact Test for Count Data
##
## data:  cleaned_data$treatment and cleaned_data$Gender_cleaned
## p-value = 1.532e-11
## alternative hypothesis: two.sided

##
## Shapiro-Wilk normality test
##
## data:  Age_cleaned[treatment == "Yes"]
## W = 0.94782, p-value = 3.913e-14

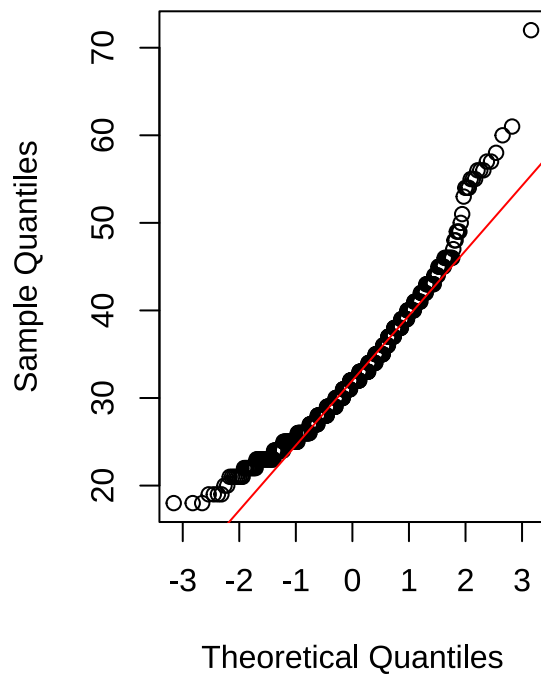
##
```



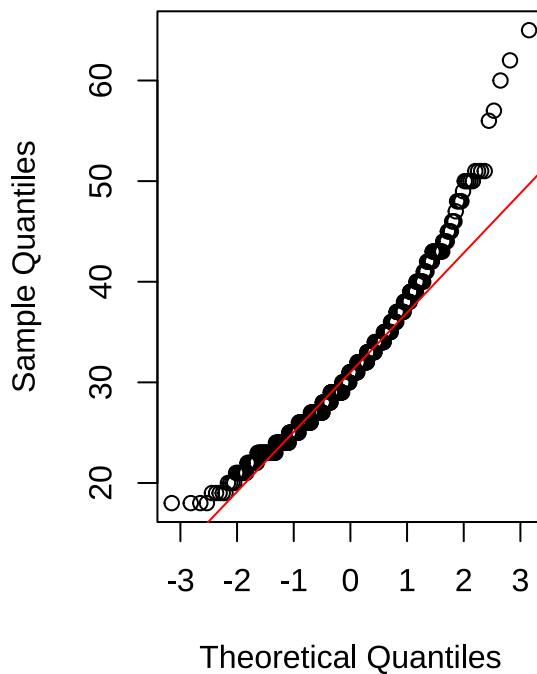
```
## Shapiro-Wilk normality test
##
## data: Age_cleaned[treatment == "No"]
## W = 0.95002, p-value = 1.272e-13

##
## F test to compare two variances
##
## data: cleaned_data$Age_cleaned by cleaned_data$treatment
## F = 0.84953, num df = 618, denom df = 631, p-value = 0.04189
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.7261393 0.9940279
## sample estimates:
## ratio of variances
##      0.8495345
```

Q-Q Plot: Treatment (Yes)



Q-Q Plot: Treatment (No)



```
##
## Wilcoxon rank sum test with continuity correction
##
## data: cleaned_data$Age_cleaned by cleaned_data$treatment
## W = 179950, p-value = 0.01417
## alternative hypothesis: true location shift is not equal to 0
```

Logistic Regression Raw Output

```
##
## Call:
## glm(formula = treatment ~ age_centred + Gender_cleaned + self_employed +
##     family_history + work_interfere_cleaned + no_employees +
##     remote_work + tech_company + benefits + care_options + wellness_program +
##     seek_help + anonymity + leave + mental_health_consequence +
##     phys_health_consequence + coworkers + supervisor + mental_health_interview +
##     phys_health_interview + mental_vs_physical + obs_consequence,
##     family = binomial, data = cleaned_data_v2)
##
## Coefficients:
##                                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)                    -2.425776    0.671649  -3.612 0.000304 ***
## age_centred                     0.029575    0.012652   2.337 0.019415 *
## Gender_cleanedMale              -0.760678    0.240063  -3.169 0.001531 **
## Gender_cleanedOther             0.130151    1.210930   0.107 0.914408
## self_employedYes               -0.190702    0.330449  -0.577 0.563872
## family_historyYes               0.946816    0.184588   5.129 2.91e-07 ***
## work_interfere_cleanedNot applicable -2.424871    0.559376  -4.335 1.46e-05 ***
## work_interfere_cleanedOften     3.513113    0.357411   9.829 < 2e-16 ***
## work_interfere_cleanedRarely    2.530646    0.297567   8.504 < 2e-16 ***
## work_interfere_cleanedSometimes  2.901663    0.264681  10.963 < 2e-16 ***
## no_employees6-25                0.165948    0.332628   0.499 0.617849
## no_employees26-100              0.348278    0.354704   0.982 0.326157
## no_employees100-500             0.289384    0.392920   0.736 0.461428
## no_employees500-1000            0.291759    0.547608   0.533 0.594181
## no_employeesMore than 1000      0.045087    0.392354   0.115 0.908513
## remote_workYes                 -0.138394    0.207538  -0.667 0.504878
## tech_companyYes                -0.058955    0.241644  -0.244 0.807251
## benefitsNo                     0.178174    0.270017   0.660 0.509341
## benefitsYes                    0.689565    0.270885   2.546 0.010909 *
## care_optionsNot sure           -0.220995    0.240707  -0.918 0.358562
## care_optionsYes                0.708006    0.240641   2.942 0.003259 **
## wellness_programNo             0.011516    0.310543   0.037 0.970419
## wellness_programYes            -0.330855    0.364452  -0.908 0.363976
## seek_helpNo                    -0.629477    0.262001  -2.403 0.016280 *
## seek_helpYes                   -0.812989    0.325190  -2.500 0.012418 *
## anonymityNo                    -0.133622    0.401513  -0.333 0.739288
## anonymityYes                   0.578368    0.237420   2.436 0.014848 *
## leaveSomewhat difficult         0.482346    0.320077   1.507 0.131819
## leaveSomewhat easy             -0.295973    0.234092  -1.264 0.206107
## leaveVery difficult            0.269747    0.350566   0.769 0.441621
## leaveVery easy                 0.091967    0.287478   0.320 0.749036
## mental_health_consequenceNo    -0.176104    0.249383  -0.706 0.480088
## mental_health_consequenceYes   -0.198039    0.259529  -0.763 0.445421
## phys_health_consequenceNo       0.053895    0.236797   0.228 0.819958
## phys_health_consequenceYes     -0.092020    0.434299  -0.212 0.832199
## coworkersSome of them          0.472525    0.249252   1.896 0.057990 .
## coworkersYes                   1.275859    0.365247   3.493 0.000477 ***
## supervisorSome of them         -0.333136    0.248581  -1.340 0.180195
## supervisorYes                  -0.279001    0.288906  -0.966 0.334187
## mental_health_interviewNo       0.432064    0.289130   1.494 0.135081
```

```
## mental_health_interviewYes          1.022249    0.656990    1.556 0.119718
## phys_health_interviewNo              0.146821    0.205882    0.713 0.475763
## phys_health_interviewYes             0.482237    0.289146    1.668 0.095356 .
## mental_vs_physicalNo                 -0.005055    0.231878   -0.022 0.982608
## mental_vs_physicalYes                 -0.056140    0.245752   -0.228 0.819303
## obs_consequenceYes                   0.235089    0.263528    0.892 0.372347
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1691.23  on 1219  degrees of freedom
## Residual deviance:  864.61  on 1174  degrees of freedom
## (39 observations deleted due to missingness)
## AIC: 956.61
##
## Number of Fisher Scoring iterations: 7
```

References

1. Mental Health in Tech Survey [Internet]. [cited 2026 May 18]. Available from: <https://www.kaggle.com/datasets/osmi/mental-health-in-tech-survey>
2. Mental Health in Tech Analysis [Internet]. [cited 2026 May 18]. Available from: <https://kaggle.com/code/muhammadusmankhan0/mental-health-in-tech-analysis>
3. OSMI Mental Health in Tech Survey 2016 [Internet]. [cited 2026 May 18]. Available from: <https://www.kaggle.com/datasets/osmi/mental-health-in-tech-2016>
4. Unpaired Two-Samples T-test in R - Easy Guides - Wiki - STHDA [Internet]. [cited 2026 May 20]. Available from: <https://www.sthda.com/english/wiki/unpaired-two-samples-t-test-in-r>
5. tbl_regression() tutorial • gtsummary [Internet]. [cited 2026 May 23]. Available from: https://www.danielsjoberg.com/gtsummary/articles/tbl_regression.html
6. Lennert F. Chapter 8 Describing data sets | An(other) introduction to R [Internet]. [cited 2026 May 23]. Available from: https://bookdown.org/f_lennert/introduction-to-r/descriptives.html
7. Plotting Estimates (Fixed Effects) of Regression Models [Internet]. [cited 2026 May 23]. Available from: https://strengjacke.github.io/sjPlot/articles/plot_model_estimates.html